

Bayesian Hierarchical MPT Modeling Theory

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Bayesian Hierarchical MPT Models

- 1] MPT models & heterogeneity
- 2] Hierarchical MPT models
- 3] Bayesian estimation with MCMC sampling
- 4] Advantages of MCMC
- 5] Application: Linking covariates to MPT parameters

MPT models & heterogeneity

Limitations of traditional MPT models (Smith and Batchelder, 2008)

- Level of analysis: Frequencies are aggregated (summed) across persons and items
- Statistical assumption: Fixed-effects model
 - ... all people behave identically
 - ... all items are similarly difficult
 - Observations are “independent and identically” (i.i.d.) distributed
 - The likelihood of all observations $i = 1, \dots, n$ is the product of the likelihood of a single observation x_i

$$p(x_1, \dots, x_n \mid \theta) = \prod_{i=1}^n p(x_i \mid \theta)$$

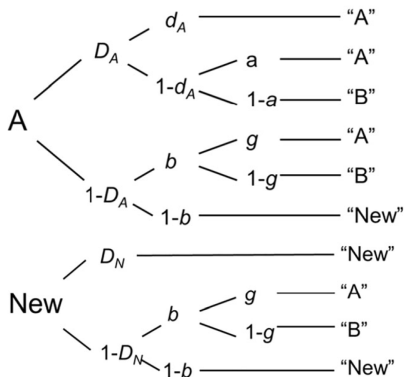
Application in psychology

- What about real data?
- What if we are interested in differences between people?

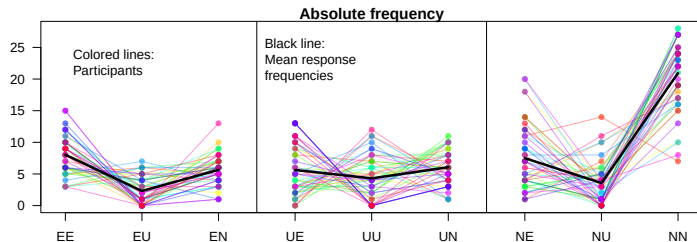
Source-Monitoring Model

Source-Monitoring

- 1) Study phase: List of words from Source A and B.
- 2) Test phase: Is the presented item from Source A/B/New?



People Behave Differently



Heterogeneity of participants

- Response frequencies are often aggregated across subjects
 - Substantial variance in the choice patterns of participants
 - Differences in memory? Response bias?
- If we fit standard MPT models to aggregated data, these differences are ignored (treated as random, unsystematic noise)
- This can result in biased statistical inferences
 - Biased point estimates if parameter are correlated
 - Over-/underestimation of confidence intervals
 - Inflated model-fit statistics

How to Handle Heterogeneity?

Three strategies (Singmann et al., 2024)

A Complete pooling:

- Analysis of aggregated frequencies
- Ignores differences between persons
- High power, but possibly biased statistical inference

B No pooling:

- A separate MPT model per person
- Low power, parameter estimates will have a large variance
- Often, not enough data per participant
- Problem: How to aggregate results across models?

C Partial pooling:

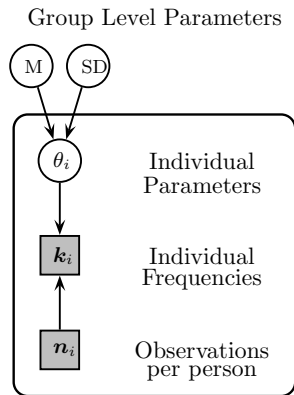
- Hierarchical/mixed modeling
- Account for differences AND similarities between persons jointly
- Higher efficiency than separate analysis
- Individual and group-level parameters inform each other

Note: This classification is very general and not limited to MPT models.

Hierarchical MPT models

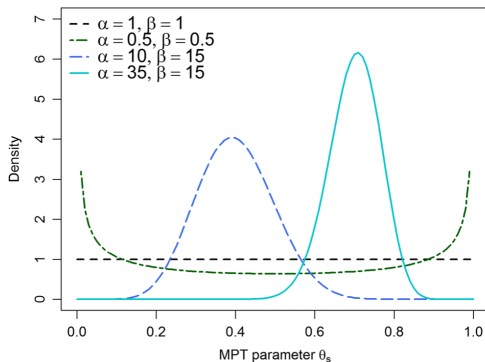
Bayesian hierarchical MPT

- Explicit assumptions about participant heterogeneity:
- MPT structure holds for each person, but with different parameters!
- One parameter vector $\theta_i = (D_i, d_i, g_i, \dots)$ per person
- At the group level, person parameters θ_i have a specific distribution
 - a. Beta-MPT
 - b. Latent-trait MPT



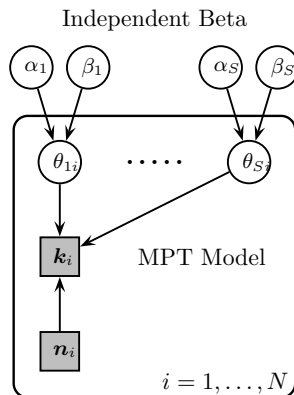
Beta distribution

- Ideally suited to model the distribution of an MPT parameter:
 - Allows values between 0 and 1
 - Two shape parameters: α and β
- On the group level, the mean for the MPT parameter equals: $\alpha/(\alpha + \beta)$



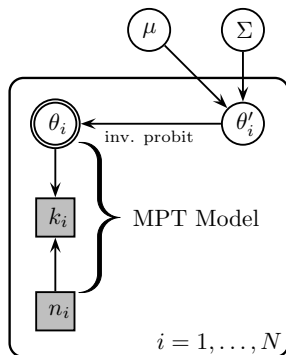
Beta-MPT (Smith and Batchelder, 2010)

- **Group level:**
 - person parameters follow independent beta distributions
 - Shape parameters α_s and β_s
- **Individual level:**
 - MPT model with separate parameters per person
 - MPT parameters: θ_{si} of person i
- **Notation for data:**
 - k_i : Individual choice frequencies
 - n_i : Number responses per person
- **Prior distribution:**
 - Uniform or gamma on α_s and β_s
 - Truncation to $\alpha_s \geq 1$ and $\beta_s \geq 1$: Unimodal group-level distribution



Latent-trait MPT model (Klauer, 2010)

- **Group level:**
 - person parameters follow a multivariate normal distribution
 - Mean μ
 - Covariance matrix Σ
- **Individual level:**
 - MPT model with separate parameters θ_i per person
 - probit-values are transformed to probabilities
 - Probit-scaled parameters: $\theta'_i = \Phi^{-1}(\theta_{si})$
- **Notation for data:**
 - k_i : Individual choice frequencies
 - n_i : Number responses per person
- **Prior distributions:**
 - Standard normal distributions for μ
 - Scaled inverse-Wishart prior for Σ



The Probit-Transformation

Transformation of MPT parameters: probability vs. probit

- We need to transform the probability parameters ($d, g, \dots$)
- We want parameters between $(-\infty, +\infty)$ (to work with normal distributions)
- Solution: Transform parameters using the cumulative density function Φ of the standard-normal distribution (similar as in logistic regression)

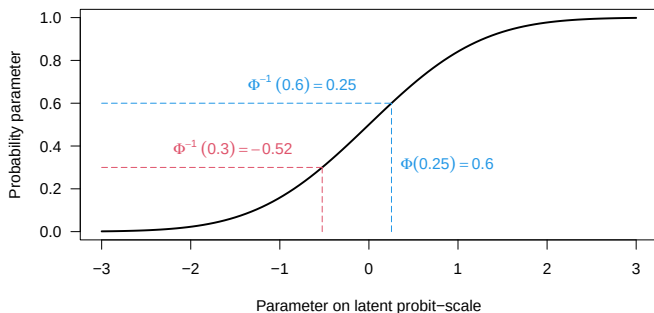


Illustration: Separate MPT Structure for each Person

Example: 2HTM for two persons

- Probit scores for memory parameter d are: $-.10$ and 1.20
- What is the predicted probability of correct OLD responses (hits)?
- We assume symmetric and identical guessing for everybody ($g = .50$)

■ Person 1:

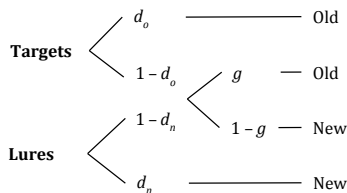
1] Transform: $d = \Phi(-.10) = .46$

2] MPT: $P(hit) = d + (1 - d)g = .46 + (1 - .46).50 = .73$

■ Person 1:

1] Transform: $d = \Phi(1.20) = .88$

2] MPT: $P(hit) = d + (1 - d)g = .88 + (1 - .88).50 = .94$

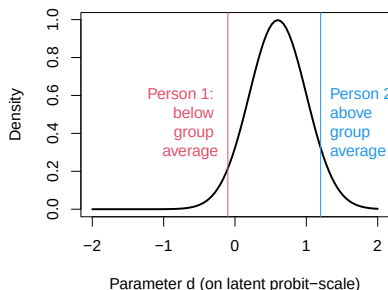


Group Level: Normal Distribution

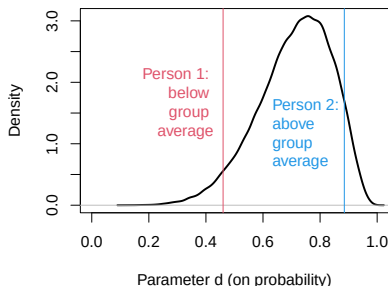
Assumption: Normal distribution of probit parameters

- Illustration: Normal distribution with mean $\mu_d = .80$ and standard deviation $\sigma_d = .3$
- For interpretation, it matters whether parameters are on the probit or the probability scale

Group-Level Distribution (latent probit)



Group-Level Distribution (probability)

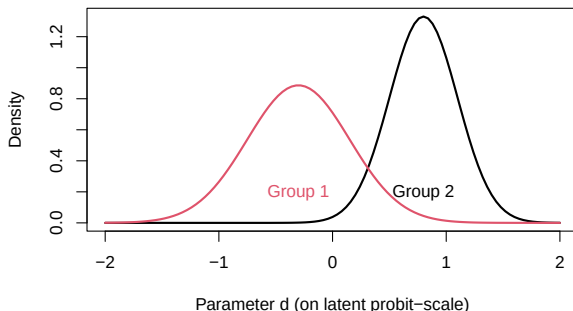


Group Level: Normal Distribution

Comparison of Groups

- Common research question: Does the group distribution differ across samples or experimental conditions?
- Example: Reduced memory for Group 1 (alcoholics) vs. Group 2 (controls)
- Similar as a simple t-test, but now we test theories about latent, person-specific probit parameters

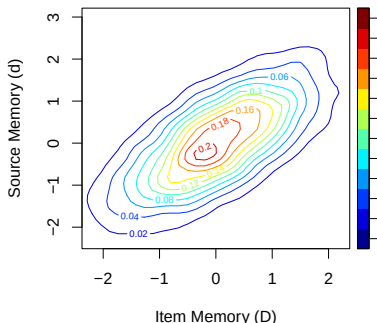
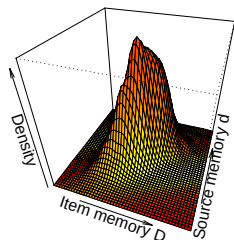
Group-Level Distribution of d



Multivariate Normal Distribution

Parameter correlations

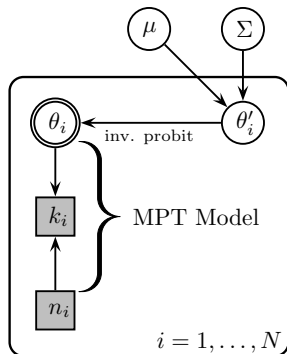
- Item and source memory might be correlated (parameters g and d)
- “The more likely I remember the item, the more likely I also remember the source.”
- Solution: Assumption that the vector θ'_i with probit-transformed MPT parameters follows a *multivariate* normal distribution
- Caveat: Correlation estimates are often very unprecise and require both large number of responses and large number of participants



Summary: Hierarchical Models

Core ideas of hierarchical models

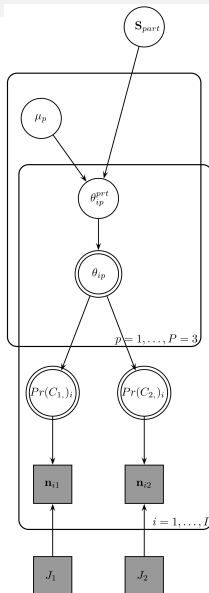
- Assume an MPT model with separate MPT parameters θ_i per person
- On the group-level, the parameters have a specific distribution
 - 1] Beta-MPT: Beta distribution
 - 2] Latent-trait MPT: multivariate normal distribution of probit-parameters with mean μ and covariance matrix Σ
 - 3] Other option (not discussed here): Discrete latent classes (Klauer, 2006)



Excursion: Graphical Models

Bayesian graphical models

- In publications, graphical models look more difficult
- Example: Matzke et al. (2015)
- However, most models use exactly the same ingredients



$$S_{part} \sim \text{Scaled-Inverse-Wishart}(\mathbf{W}, df = P + 1, \xi_{part})$$

$$\xi_{part_p} \sim \text{Uniform}(0, 100)$$

$$\mu_p \sim \text{Normal}(0, 1)$$

$$\theta_i^{prt} \sim \text{Multivariate-Normal}\left((\mu_1, \dots, \mu_P), S_{part}^{-1}\right)$$

$$\theta_{ip} = \phi(\theta_{ip}^{prt})$$

$$Pr(C_{11})_i = \theta_{i1} \times \theta_{i2}$$

$$Pr(C_{12})_i = (1 - \theta_{i1}) \times \theta_{i3}^2$$

$$Pr(C_{13})_i = (1 - \theta_{i1}) \times 2 \times \theta_{i3} \times (1 - \theta_{i3})$$

$$Pr(C_{14})_i = \theta_{i1} \times (1 - \theta_{i2}) + (1 - \theta_{i1}) \times (1 - \theta_{i3})^2$$

$$Pr(C_{21})_i = \theta_{i3}$$

$$Pr(C_{22})_i = (1 - \theta_{i3})$$

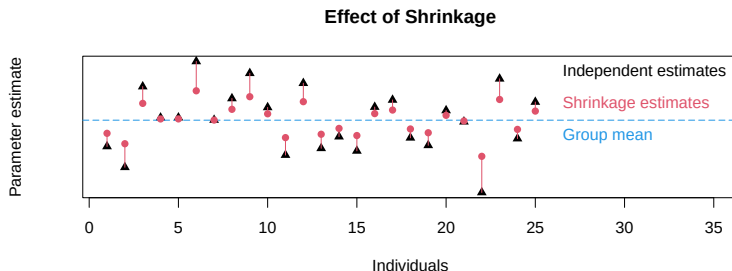
$$\mathbf{n}_{i1} \sim \text{Multinomial}(Pr(C_{1,})_i, J_1)$$

$$\mathbf{n}_{i2} \sim \text{Multinomial}(Pr(C_{2,})_i, J_2)$$

Some Advantages

Benefits of hierarchical MPT models

- Avoid aggregation biases: We get estimates for each person
- “Shrinkage” of parameter estimates
 - Parameter estimates for each person are closer together compared to fitting each person separately
 - Hence, extreme estimates are less likely
 - Overall, this ensures that parameter estimates are closer to the true values on average
- The basic idea of hierarchical models can easily applied to any other model
 - 1] Assume that model holds for each person
 - 2] Specify group-level distribution of parameters across persons



Bayesian estimation with MCMC

Parameter estimation

- How can we actually fit such models?
- Which are the “best” parameters given the data?
 - Standard MPT models: Maximum likelihood estimation
 - Not an option for hierarchical models (intractable likelihood function due to high-dimensional integrals)

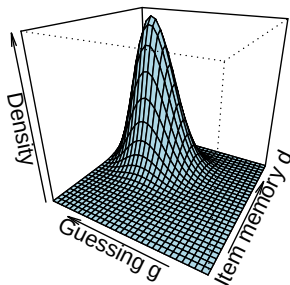
Solution

- Hierarchical models are often fitted using Bayesian statistics

Maximum Likelihood

- Logic of parameter estimation with maximum-likelihood
 - 1) Define likelihood function $p(x | \theta)$
 - 2) Find parameters θ that maximize f
- Interpretation: “The estimator $\hat{\theta}$ has the highest likelihood.”
- Computational solution: search for the “top of the mountain”

Likelihood



Bayesian Estimation

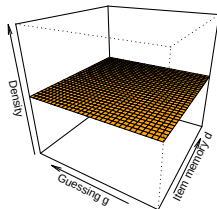
- Logic of Bayesian parameter estimation

- 1) Define likelihood $p(x | \theta)$ and prior distribution $p(\theta)$
- 2) Derive the posterior distribution of the parameters via Bayes' theorem:

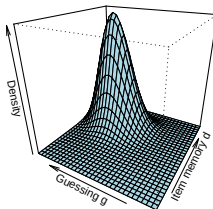
$$p(\theta | x) = \frac{p(x | \theta)p(\theta)}{p(x)}$$

- Interpretation: "What have we learned about the parameters θ given the data x ?"

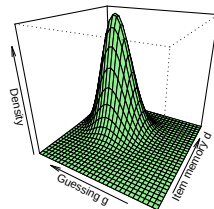
Prior



Likelihood



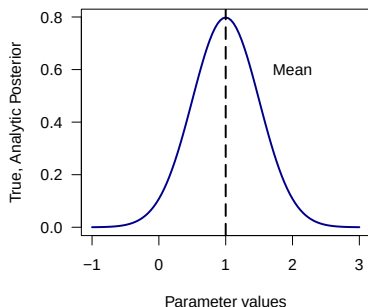
Posterior



Bayesian Estimation

- **Problem:** We need to work with the posterior function $p(\theta | x)$
 - What is the mean/mode/95% credibility interval of θ ?
 - Often, this is analytically not tractable
- **Solution:** We draw random samples from the posterior distribution
 - Logic: It is easier to draw conclusions from these random samples than deriving solutions for the analytical posterior (which is a function!)
 - Example: Computing the mean of a normal distribution requires to solve:

Target Distribution



Monte Carlo Approximation

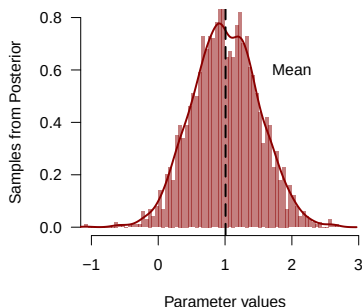
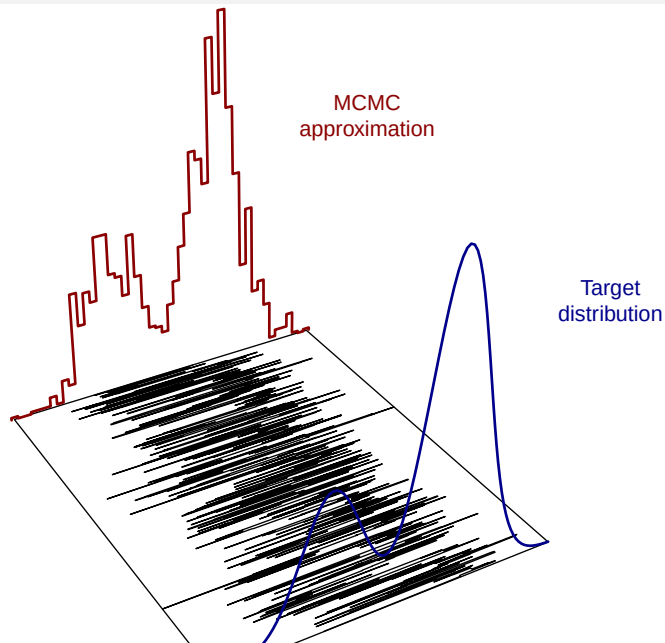


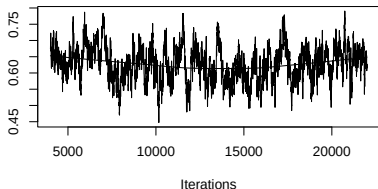
Illustration of MCMC



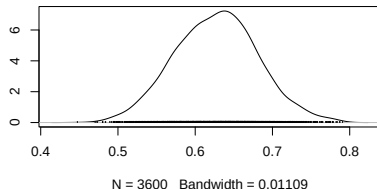
Markov Chain Monte Carlo (MCMC) Sampling

- 1 Draw random samples of the posterior distribution for *all* parameters (individual and group level)
- 2 Summarize parameter samples (e.g., mean, SD, density, ...)

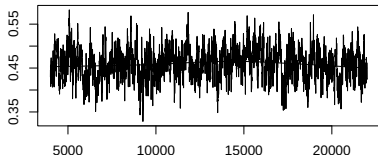
Trace of mean[dn]



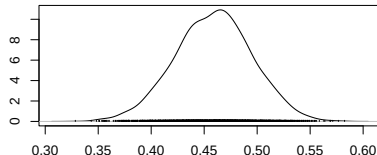
Density of mean[dn]



Trace of mean[g]



Density of mean[g]



Markov chain Monte Carlo (MCMC)

- General method to draw posterior samples
- In a hierarchical model, there are many (!) parameters
 - Group-level means and covariances, person parameters, ...
 - Intuitively, this method moves around and searches for parameter values with high posterior density
- There are software packages that draw random samples for many models of interest
 - JAGS, WinBUGS, OpenBUGS, Stan, ...

Summary of Bayesian estimation

- 1) Develop a model ($\Rightarrow$ psychological theory, multiTree)
- 2) Get posterior (MCMC) samples (JAGS, TreeBUGS)
- 3) Summarize these samples (e.g., mean of group-level parameters $\mu_D, \mu_g, \dots$)

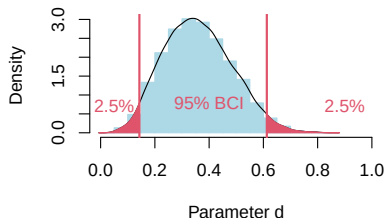
Advantages of MCMC

Advantages of MCMC: Uncertainty

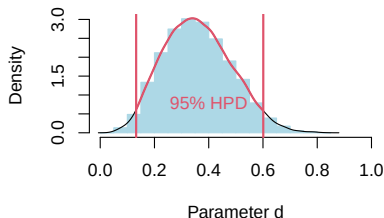
Advantages of MCMC sampling

- **Theoretical:** No asymptotic assumptions
 - Maximum likelihood: requires a sufficient number of observations
- **Practical:** It is easy to quantify uncertainty
 - Bayesian credibility interval (BCI): What are the 2.5%- and 97.5%-quantiles of the parameter values?
 - Highest posterior density interval (HPD or HDI): What are the 95% most plausible parameter values?
 - For probability parameters, these intervals will always be in the interval $[0, 1]$

Bayesian Credibility Interval



Highest Posterior Density Interval

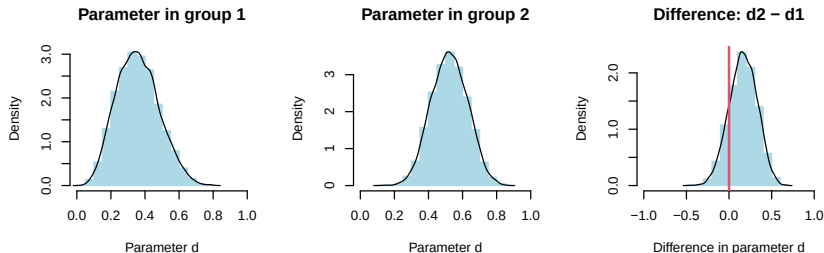


Advantages of MCMC: Transformed Parameters

- Often, we are interested in parameter/group comparisons
 - Example: Do healthy controls vs. schizophrenics differ in memory?
 - Test: Does the group-mean parameter μ_D differ?
- Based on MCMC samples, we can directly estimate functions of the parameters

MCMC estimation of transformed parameters

- 1 Draw MCMC samples
- 2 Compute transformed parameters for all samples
 - Example: $\delta^{(t)} = \theta_1^{(t)} - \theta_2^{(t)}$
- 3 Summarize the new values

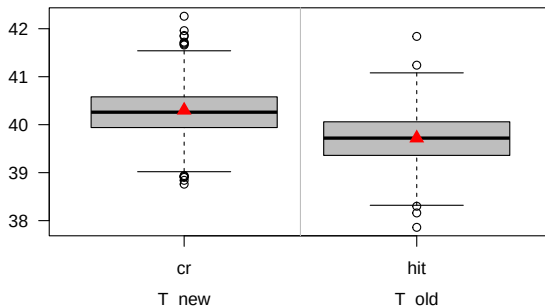


Advantages of MCMC: Model Fit

Does the model fit the data?

- Graphical comparison: observed vs. predicted frequencies
 - Use posterior samples of the MPT parameters to sample new data (= posterior predictive)
 - Compare whether these predicted data (boxplot) are in line with the observations (red points)

Observed (red) and predicted (boxplot) mean frequencies



How to quantify model fit for MPT models?

- Test statistic similar to Pearson's χ^2 statistic (Klauer, 2010)
 - T1 statistic: Mean structure of frequencies
 - T2 statistic: Covariance matrix of frequencies
- Posterior predictive p -value (PPP) measures model fit:
 - 1) Compute T1 for the observed data
 - 2) Compute T1 for the posterior predicted data
 - 3) PPP = probability that T1(predicted) is larger than T1(observed)
- Ideally, PPP should be around .50

Application

Linking Covariates to MPT Parameters

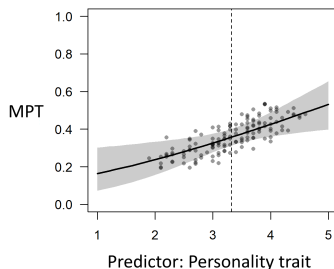
Interindividual differences

- Personality as a predictor for MPT parameters
- Statistical approach in latent-trait MPT: Similar to logistic regression

$$\theta_i = \Phi(\mu + \boxed{\beta \cdot x_i} + \delta_i)$$

Cognitive Psychometrics (Riefer et al., 2002)

- Using cognitive (MPT) models to learn about interindividual differences



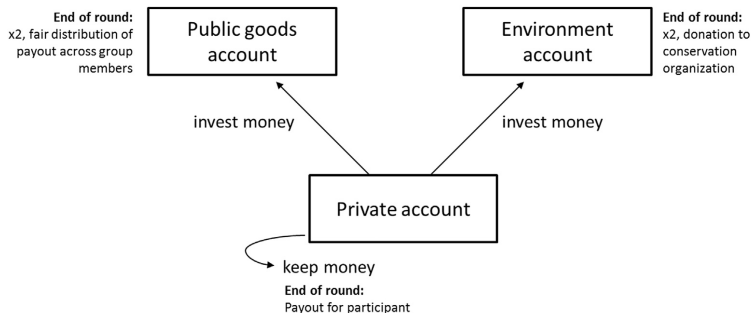
Example: Linking personality to MPT parameters

- “Which is the greater good? A social dilemma paradigm disentangling environmentalism and cooperation”
 - Klein et al. (2017) (*Journal of Environmental Psychology*)
- Research question: How can we distinguish between 3 types of behavior?
 - Pro-environmental behavior
 - Pro-social behavior
 - Selfish behavior



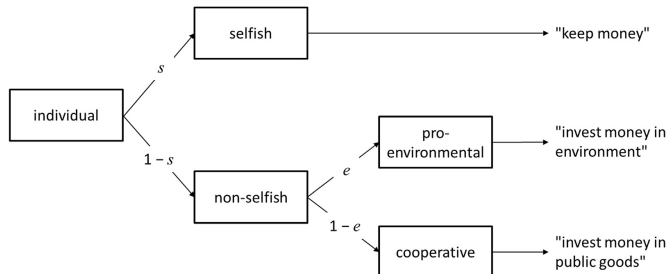
The Greater Good Game (Klein et al., 2017)

- Participants decide whether to keep the money for themselves or contribute it to either a public goods or an environment account.
- Important: Participants are forced to decide between the group and the environment!
- The game is a variant of a nested public goods game



MPT model for the Greater Good Game

- s = probability of selfish behavior
- e = probability of pro-environmental behavior



Results

- Honesty Humility (= sincerity, fairness) is associated with less selfish behavior
- Selfish behavior decreases from 33.4% to 13.9% for participants $-1/ +1$ SD on Honesty Humility

Hierarchical MPT Models

- Individual level
 - Assume a separate MPT model for each person
- Group level
 - Beta-MPT: Beta distribution of person parameters
 - Latent-trait MPT: Normal distribution of probit-transformed parameters
- Bayesian model fitting: Drawing posterior samples

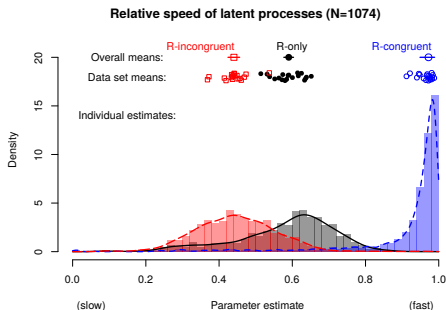
Appendix & References

Appendix A: Meta-Analysis of Raw Data

- Linking process and measurement models of recognition-based decisions (Heck and Erdfelder, 2017)
- Reanalysis of about 400,000 decisions
 - 3-level hierarchical latent-trait MPT:

$$\theta_{sij} = \Phi(\mu_s + \boxed{\xi_{sj}} + \delta_{si})$$

- Overall mean of MPT parameters (μ_s)
- Participants nested in studies (random effect: ξ_{sj})
- Responses nested in participants (random effect: δ_{si})



Appendix B: Large-Scale Reanalysis

Open questions:

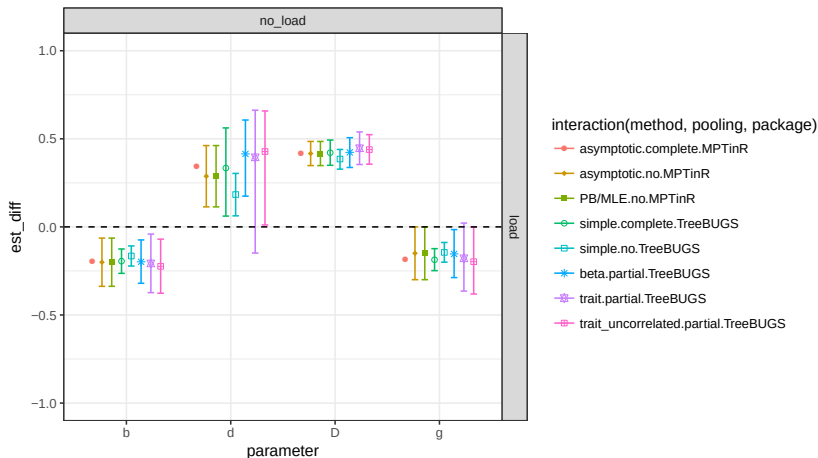
- How much do results actually differ between different MPT versions?
- Which MPT version should be used in practice?

Large-scale reanalysis project

- Network of MPT researchers (organized by Beatrice Kuhlmann & Julia Groß)
- Reanalysis of existing data sets to compare:
 - Fixed-effects vs. hierarchical
 - Maximum-likelihood vs. Bayes
 - Different hierarchical level-2 structures
(beta, multiv. normal, independent univ. normal)
- Software: “A multiverse pipeline for MPT models”
 - Maximum likelihood: `MPTinR` (Henrik Singmann)
 - Bayes: `TreeBUGS` (Heck et al., 2018)
 - Available at: <https://github.com/mpt-network/MPTmultiverse>

Appendix C: Reanalysis with Different Models

- Source-monitoring model (data by Bayen & Kuhlmann, 2011)
- Plot: Difference in parameters across two groups



References I

- Heck, D. W., Arnold, N. R., and Arnold, D. (2018). TreeBUGS: An R package for hierarchical multinomial-processing-tree modeling. *Behavior Research Methods*, 50:264–284.
- Heck, D. W. and Erdfelder, E. (2017). Linking process and measurement models of recognition-based decisions. *Psychological Review*, 124:442–471.
- Klauer, K. C. (2010). Hierarchical multinomial processing tree models: A latent-trait approach. *Psychometrika*, 75:70–98.
- Klein, S. A., Hilbig, B. E., and Heck, D. W. (2017). Which is the greater good? A social dilemma paradigm disentangling environmentalism and cooperation. *Journal of Environmental Psychology*, 53:40–49.
- Matzke, D., Dolan, C. V., Batchelder, W. H., and Wagenmakers, E.-J. (2015). Bayesian estimation of multinomial processing tree models with heterogeneity in participants and items. *Psychometrika*, 80:205–235.
- Riefer, D. M., Knapp, B. R., Batchelder, W. H., Bamber, D., and Manifold, V. (2002). Cognitive psychometrics: Assessing storage and retrieval deficits in special populations with multinomial processing tree models. *Psychological Assessment*, 14:184–201.

References II

- Singmann, H., Heck, D. W., Barth, M., Erdfelder, E., Arnold, N. R., Aust, F., Calanchini, J., Gümüşdaglı, F. E., Horn, S. S., Kellen, D., Klauer, K. C., Matzke, D., Meissner, F., Michalkiewicz, M., Schaper, M. L., Stahl, C., Kuhlmann, B. G., and Groß, J. (2024). Evaluating the robustness of parameter estimates in cognitive models: A meta-analytic review of multinomial processing tree models across the multiverse of estimation methods. *Psychological Bulletin*, 150:965–1003.
- Smith, J. B. and Batchelder, W. H. (2008). Assessing individual differences in categorical data. *Psychonomic Bulletin & Review*, 15:713–731.
- Smith, J. B. and Batchelder, W. H. (2010). Beta-MPT: Multinomial processing tree models for addressing individual differences. *Journal of Mathematical Psychology*, 54:167–183.